INCH-POUND

MIL-PRF-32816
29 June 2026

PERFORMANCE SPECIFICATION

NICKEL ALUMINUM BRONZE, WIRE ARC, DIRECTED ENERGY DEPOSITION, ADDITIVELY MANUFACTURED

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers additively manufactured (AM) nickel aluminum bronze material using the wire arc directed energy deposition (DED) process.

1.2 Classification. Material is of the following feedstock grades and classes, as specified (see 6.2):

1.2.1 Grades. (see 4.8)

a. Grade A – Material manufactured in compliance with S9074-A4-GIB-010/AM-WIRE DED.

b. Grade B – Material manufactured in compliance with manufacturer specifications or industry standards (see 6.3).

1.2.2 Classes.

a. Class 1 – Material manufactured using wire feedstock that is in compliance with MIL-DTL-23765/3, type MIL-CuNiAl.

b. Class 2 – Material manufactured using wire feedstock that is in compliance with AWS A5.7/A5.7M, ERCuNiAl.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-23765/3 - Electrodes and Rods – Welding, Bare, Solid, Copper Alloy

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-792 - Manufacturing Process Identification

MIL-STD-1684 - Control of Heat Treatment

(Copies of these publications are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-A4-GIB-010/AM-WIRE DED	-	Requirements for Metal Directed Energy Deposition Additive Manufacturing
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). These publications is/are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS A5.7/A5.7M	-	Specification for Copper and Copper-Alloy Bare Welding Rods and Electrodes

(Copies of this publication are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM E8/E8M	-	Standard Test Methods for Tension Testing of Metallic Materials
ASTM E478	-	Standard Test Methods for Chemical Analysis of Copper Alloys

(Copies of these publications are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS2750	-	Pyrometry

(Copies of this publication are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Feedstock. Wire feedstock shall be in accordance with the standard (MIL-DTL-23765/3 or AWS A5.7/A5.7M) of the appropriate class (see 1.2). Each manufacturing lot (see 4.3) shall be produced from the minimum practicable number of lots of feedstock (see 6.6.2).

3.2 Build platform. Unless otherwise specified (see 6.2), the build platform shall not be integrated into the printed material. If material other than S-36A nickel aluminum bronze, as defined in S9074-A4-GIB-010/AM-WIRE DED, is used for a non-integrated build platform, a minimum of 0.5 inch of discardable AM build material shall separate the build platform from the deposited material's attached dimensional surface. Methods used for removal of the build platform (e.g., band saw, water jet, or electro-discharge machining with recast layer removed) shall not affect the mechanical properties of the AM material.

3.3 Chemical composition. Feedstock and all DED material shall conform to the compositions specified in table I. See 4.4.1 and 4.5 for composition analysis requirements.

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TABLE I. Chemical composition, weight-percent. [1/, 2/]

	Class 1	Class 2
Copper plus silver [3/]	Balance	Balance
Zinc	*	0.10
Tin	*	*
Manganese	0.60 to 3.50	0.60 to 3.50
Iron	3.00 to 5.00	3.0 to 5.0
Silicon	*	0.10
Nickel plus cobalt [4/]	4.00 to 5.50	4.0 to 5.5
Phosphorus	*	*
Aluminum	8.50 to 9.50	8.50 to 9.50
Lead	0.020	0.02
Total other elements	0.50	0.50

NOTES:

1/ Analysis shall be made for the elements indicated by an asterisk (*) and those for which specific values are shown in this table. If, however, the presence of other elements is indicated during routine analysis, further analysis shall be made to determine their content. The total of these other elements shall not exceed the limits specified for the "total other elements" in the last row of the table.

2/ Single values are maximums.

3/ Silver may or may not be present.

4/ Cobalt may or may not be present.

3.4 Heat treatment. Material shall be heat treated in furnaces in accordance with the pyrometry requirements of SAE AMS2750, class 5 or better. Material shall be heated to 1250 °F for a minimum of 6 hours. Material shall be furnace cooled to 500 °F then air cooled to room temperature. When specified (see 6.2), heat treatment shall be in accordance with MIL-STD-1684.

3.5 Tensile properties. The tensile properties of all material shall meet or exceed the values specified in table II. Reduction of area shall be measured and reported for information.

TABLE II. Tensile properties.

Property	Minimum value
Ultimate tensile strength (ksi)	85
Yield strength, 0.2% offset (ksi)	35
Elongation in 2 inches (%)	15
Reduction of area (%)	Information only

3.6 Weld repair. Unless otherwise specified (see 6.2), weld repair shall be in accordance with S9074-AR-GIB-010/278.

3.6.1 Post-weld heat treatment on as-printed DED material. When weld repairs are made to the DED material that has not yet been subjected to heat treatment, the material shall be heat treated by the same procedure specified in 3.4 following the repair.

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3.6.2 Post-weld heat treatment on weld repaired, heat treated DED material. Unless otherwise specified (see 6.2), weld repairs made to heat-treated DED material shall not be subjected to a post-weld heat treatment (PWHT). If PWHT is performed, redetermination of mechanical properties is required following PWHT.

3.7 Additional mechanical properties. Additional mechanical properties shall be as specified (see 6.2).

3.8 Surface condition. Unless otherwise specified (see 6.2), material shall be ground, machined, or otherwise finished to produce smooth surfaces.

3.9 Detrimental material controls. During the manufacturing processes, examinations, and tests, the material shall not come into direct contact with mercury or any of its compounds, nor with any mercury-containing device employing a single boundary of containment. Additional detrimental material controls shall be as specified (see 6.2).

3.10 Identification marking. Material other than specimens used for conformance inspections shall be marked with the number and revision of this specification, manufacturer name or CAGE code, and manufacturing lot number. Markings shall be permanently applied in accordance with MIL-STD-792; an alternate marking method is permissible, provided it is an available option in the contractually invoked drawing or specification. Markings shall be located so as not to affect the form, fit, or function of the material and must be legible after all required surface finishing and post-processing operations.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.2).

4.2 Conformance inspection. Conformance inspection shall include the examinations and tests of table III.

TABLE III. Conformance inspections.

Examinations and tests	Requirement	Conformance inspection	Sampling requirements
Feedstock	3.1	4.8	---
Build platform	3.2	4.8	---
Chemical composition	3.3	4.5 and 4.8	4.4.1 and 4.4.3
Heat treatment	3.4	4.8	---
Tensile properties	3.5	4.6 and 4.8	4.4.2 and 4.4.3
Weld repair	3.6, 3.6.1, and 3.6.2	4.8	---
Additional mechanical properties	3.7	4.7 and 4.8	---
Surface condition	3.8	4.8	---
Detrimental material controls	3.9	4.8	---
Identification marking	3.10	4.8	---

4.3 Lot size. For the purposes of inspections and tests, material produced in one build cycle, as defined in S9074-A4-GIB-010/AM-WIRE DED, and heat treated in the same furnace load shall constitute a manufacturing lot.

4.4 Sampling.

4.4.1 Chemical composition. A minimum of one sample shall be tested from each lot in its final heated condition.

4.4.2 Tensile properties. A minimum of two tensile specimens shall be tested from each lot in its final heat treated condition. Tensile specimens shall be machined from bulk deposition or near-net shape blanks. Tensile specimens shall conform to the ASTM E8/E8M standard round specimen size 1 geometry with a gauge length four

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times the diameter. At least one specimen shall be oriented with its length in the Z-axis and at least one specimen shall be oriented with its length in the XY plane (see 6.6.1).

4.4.3 Resampling and retesting. Unless otherwise specified (see 6.2), non-conforming test results may be addressed by testing two additional samples taken from the same manufacturing lot as the original non-conforming sample. The average of all test results shall meet specified requirements for lot acceptance. If any two samples of any type of test fail to meet the acceptance criteria, then the manufacturing lot shall be rejected. When specified (see 6.2), any manufacturing lot may be rejected based on a single tensile test.

4.5 Chemical composition analysis. The chemical composition of each lot shall be tested and analyzed by accepted analytical methods. In case of disagreements, the referee method shall be ASTM E478. The chemical composition of each lot shall conform to the compositions shown in table I.

4.6 Tensile testing. Tensile specimens shall be tested in accordance with ASTM E8/E8M and shall meet the requirements of table II (see 3.5).

4.7 Additional mechanical properties. Test methods and sampling requirements for additional mechanical testing shall be as specified (see 6.2).

4.8 Certificate of conformance. A certificate of conformance (COC) shall be provided for each lot of material offered for acceptance in accordance with the lot definition specified in 4.3. The COC shall include the following:

a. Statement that each lot has been sampled, tested, and inspected in accordance with the requirements specified herein.

b. For grade A material (see 1.2.1), record of approval from the Technical Authority (see 6.6.3) or authorized representative that indicates the manufacturer is in compliance with S9074-A4-GIB-010/AM-WIRE DED.

c. For grade B material (see 1.2.1), statement specifying the manufacturer specification or industry standard(s) used to determine material compliance and a record of approval from the Technical Authority (see 6.6.3) or authorized representative thereof.

d. Record of the build platform including thickness and COC from manufacturer (see 3.2).

e. Complete record of all heat treatments, including time, temperatures, atmosphere, and heating and cooling rates (see 3.4).

f. Statement whether a weld repair has been performed (see 3.6), including complete record of all repaired defects with location, weld repair inspection results, post-weld heat treatment (if any) (see 3.6.1 and 3.6.2), and associated weld procedure approvals.

g. Statement describing surface finishing method(s) used (see 3.8).

h. Statement certifying that detrimental material controls have been met been met (see 3.9).

i. Statement certifying that all material markings comply with the requirements of this specification (see 3.10).

j. Complete record of testing (see table III) used to demonstrate compliance with this specification, including any failed tests.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Wire arc DED nickel aluminum bronze AM material is intended as an alternative for castings specified in MIL-B-24480, “Bronze, Nickel-Aluminum (UNS No. C95800) Casting for Seawater Service,” and ASTM B148, “Standard Specification for Aluminum-Bronze Sand Castings,” for use in Navy applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grade and class required (see 1.2).

c. Permissible material specification and grade for integrated build platform and any additional test requirements associated with its use (see 3.2).

d. When heat treatment in accordance with MIL-STD-1684 is required (see 3.4).

e. Weld repair requirements or restrictions (see 3.6).

f. Additional mechanical properties and corresponding acceptance criteria (see 3.7).

g. When surface finishing is not required or is other than specified (see 3.8 and 6.4).

h. When needed, additional detrimental material control requirements (see 3.9).

i. Requirements for resampling and retesting (see 4.4.3).

j. Test specimen number, geometry, orientation, and test method requirements for supplemental mechanical testing (see 4.7).

k. Packaging requirements (see 5.1 and 6.5).

6.3 DED qualification. Grade B material requires the disclosure of a process qualification standard. Examples of appropriate commercial standards include SAE AMS7005, “Wire Fed Plasma Arc Directed Energy Deposition Additive Manufacturing Process,” and AWS D20.1/D20.1M, “Specification for Fabrication of Metal Components Using Additive Manufacturing.” These documents provide a framework to prove that the DED process is capable of producing material of consistent quality. Use of grade A material is recommended for critical applications.

6.4 Surface finish. Material produced using DED may have a rough surface finish that may be associated with a reduction in corrosion resistance and fatigue life. A rough surface may also reduce inspectability by restricting the use of NDT methods such as ultrasonic testing and dye penetrant testing, particularly when average roughness exceeds 250 microinch. Material is often post-processed via machining, grinding, electrical discharge machining, or polishing to achieve desired surface finish. To ensure good performance, prescriptive surface finishing requirements should be incorporated into acquisition requirements, as specified (see 6.2).

6.5 Packaging marking. In addition to other packaging and labeling requirements required by the contract (see 6.2), each unit, intermediate, and exterior shipping container should be marked with the material specification number, manufacturer’s name or CAGE code, and manufacturing lot number. This marking should be placed on each MIL-STD-129, “Military Marking for Shipment and Storage,” label required by the contract and should be placed either below the identification test marking or in a conspicuous location on the identification marked side of the container.

6.6 Definitions.

6.6.1 DED build orientations. The Z orientation is parallel to the layer-by-layer build direction (i.e., perpendicular to the surface of the build platform if no reorientation has occurred). The X-axis is parallel to the weld axis, and the Y-axis is perpendicular to the weld axis and parallel to the layer plane.

6.6.2 Feedstock lots. Feedstock lots are defined in accordance with AWS A5.01M/A5.01, “Welding and Brazing Consumables – Procurement of Filler Metals and Fluxes.”

6.6.3 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05T may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

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6.7 Subject term (key word) listing.

3-D Printing
Casting Substitute
Metal Printing

CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – SH

Preparing activity:
Navy – SH
(Project AMPR-2026-007)

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